

ACTUARIAL MEMORANDUM

To:	Mortgage Risk Unit and Underwriting Department
From:	Actuarial / Analytics Team
Subject:	Use of the House-Price AVM for Partially Finished Properties in Madokero and Mabvazuva
Date:	24 September 2026

The final Random Forest automated valuation model (AVM) produces a point estimate of US\$95,349.07 for the specified Madokero property (House A) and US\$94,706.26 for the otherwise comparable Mabvazuva property (House B). The corresponding 90% split-conformal prediction intervals are US\$17,578.78–US\$517,182.89 for House A and US\$17,460.27–US\$513,696.23 for House B.

The predicted difference between the two properties is therefore only US\$642.81, with House A receiving the slightly higher model valuation. The very wide and substantially overlapping prediction intervals indicate considerable valuation uncertainty and mean that the small difference between the two point estimates should not be interpreted as a precise market differential.

(a) Location loading and its use in lending/underwriting

The model's counterfactual analysis indicates only a small sensitivity to location and title status for these particular synthetic properties. Holding title status constant, changing the location from Madokero/Harare West to Mabvazuva/Ruwa changes the predicted value by approximately –US\$192.09. Similarly, changing the tenure indicator from title deeds to cession produces a model sensitivity of approximately –US\$187.62 in Madokero and –US\$187.24 in Mabvazuva.

These values should not be interpreted as universal market discounts or causal effects. They represent local sensitivities of the fitted model for the specified property characteristics.

There is evidence that location is important more generally. In the formal statistical analysis, suburb remained significantly associated with log advertised price after controlling for floor size and bedrooms ($F = 17.13$, $p < 0.001$; partial $\eta^2 = 0.781$). The feature-ablation analysis also showed that removing location increased test log-RMSE from 0.6278 to 0.6541, indicating that geographic information contributes to out-of-sample predictive performance.

However, the case-study location differential should not be applied mechanically as a lending or underwriting loading. The density diagnostic gives House A a 5-nearest-neighbour distance of 2.3233, corresponding to approximately the 27th percentile of the training-distance distribution and placing it within a relatively common training region. House B has a distance of 2.6066, approximately the 71st percentile, indicating a moderately sparse region.

Although four of House B's five nearest training observations are actual Mabvazuva listings, the uncertainty intervals remain extremely wide. The AVM's location loading is therefore useful as supporting evidence but is not sufficiently precise to constitute a stand-alone adjustment for collateral valuation.

(b) Implications for mortgage LTV limits and homeowners' sums insured

The case study demonstrates an important limitation of using an AVM for partially finished residential properties. Both houses are structurally complete but remain unfinished in several respects: they are unpainted internally and externally, have partial ceilings, unfitted kitchens, bare or screeded cement floors, unpaved yards and no perimeter fencing. They have municipal water and ZESA electricity but no solar system.

The model incorporates condition and finish information through text-derived indicators and CLIP-based visual scores. However, no photographs were supplied for the two synthetic case-study properties. Consequently, their visual features were represented using training-sample median values. Floor size was also not specified in the case-study description and was therefore represented using the training median. These assumptions increase uncertainty around the resulting valuations.

Accordingly, the US\$95,349.07 and US\$94,706.26 point estimates should not be treated as exact market values for setting mortgage loan-to-value limits. The wide conformal intervals demonstrate that plausible values under the fitted model cover a much broader range. For lending purposes, the AVM should therefore operate as a screening, benchmarking or decision-support tool, supplemented by a professional physical valuation where the property is unfinished or otherwise atypical.

The same distinction is important for homeowners' insurance. A house-price AVM estimates market value rather than the cost of rebuilding the property. Consequently, it should not be used directly to determine the sum insured. Reinstatement or reconstruction cost should instead be assessed separately using appropriate building-cost information, particularly where construction remains incomplete.

The small modelled title-versus-cession effect should likewise not be interpreted as evidence that title deeds and developer cession have equivalent economic or legal implications. It represents the behaviour of this fitted model within the available sample rather than a general valuation rule.

(c) Recommended methodological improvement

The priority methodological improvement is to expand the training dataset with additional unfinished and partially finished properties across Madokero, Mabvazuva and other Greater Harare suburbs, while recording condition characteristics explicitly.

Future data collection should separately identify internal and external painting, ceilings, flooring, fitted kitchens, paving, perimeter walls and fencing, utilities, solar installations, completion percentage and precise tenure status. This would reduce dependence on sparse description language and allow the model to distinguish more reliably between location, structural characteristics and level of finish.

A larger dataset would also permit repeated spatial cross-validation and more reliable uncertainty calibration. Where possible, actual transaction prices should supplement advertised prices because asking prices may differ from realised market values.

Conclusion

The AVM provides useful indicative values for the two properties, but the wide prediction intervals and incomplete representation of their finish characteristics mean that the estimates should be used as decision-support information rather than stand-alone professional valuations. For mortgage LTV decisions, an independent property valuation remains appropriate for atypical or partially finished

properties, while homeowners' sums insured should be based on reconstruction or reinstatement cost rather than AVM market value.